

Projeto do Pilar P7

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$$14 \times 30 = 420 \text{ cm}^2$$

Esforço normal

Reações dos vigas: V7: 35,01 kN
V21: 172,56 kN

$$A_c = \frac{1,4 N_d}{0,6 \cdot 2,5 + 0,42} = \frac{N_d}{1,37}$$

Piso	Reação (kN)	Estimativa de N_d (kN)	A_c (cm ²)	h_x (cm)	h_y (cm)
cobertura	172,56	$172,56 \cdot 1,25 = 215,7 \text{ kN}$	157,4	14	30
1º andar	$35,01 + 172,56$	$215,7 + (35,01 + 172,56) \cdot 1,25 = 475,2 \text{ kN}$	346,9	14	30
térreo	$35,01 + 172,56$	$475,2 + (35,01 + 172,56) \cdot 1,25 = 734,6 \text{ kN}$	536,2	19	30

Piso: N_d (kN)

1º andar: $215,7 + 5,8 = 221,5$

térreo: $221,5 + 259,5 + 5,8 = 486,8$

fundações: $486,8 / 1,25 + 35,01 + 172,56 + 3,59 = 600,6$

p.p.d. ($14 \times 30 \times 315$): $1,4 \cdot 1,25 \cdot 0,14 \cdot 0,30 \cdot 315 \cdot 25 = 5,79 \text{ kN} \approx 5,8 \text{ kN}$

p.p.d. ($19 \times 30 \times 180$): $1,4 \cdot 0,19 \cdot 0,30 \cdot 180 \cdot 25 = 3,59$

Momentos fletores devido à V7

$$M_{ep} = 32,09 \text{ kNm} = 3209 \text{ kNcm}$$

Pilar: 14×30 : $I = \frac{30 \cdot 14^3}{12} = 6860 \text{ cm}^4$ 19×30 : $I = \frac{30 \cdot 19^3}{12} = 17148 \text{ cm}^4$

2º liga: $I_{vig} = \frac{14 \cdot 40^3}{12} = 74667 \text{ cm}^4$

Pilar: 1º andar: $n = \frac{2 \cdot 6860}{630} = 21,78 \text{ cm}^3$

2º liga: $n = \frac{74667}{550} = 135,76 \text{ cm}^3$

térreo: $n = \frac{2 \cdot 6860}{315} = 43,56 \text{ cm}^3$

fundações: $n = \frac{2 \cdot 17148}{180} = 190,53 \text{ cm}^3$

2º liga 1º andar: $n_{sup} = 21,78 \text{ cm}^3$ $n_{inf} = 43,56 \text{ cm}^3$ $n_{vig} = 135,76 \text{ cm}^3$

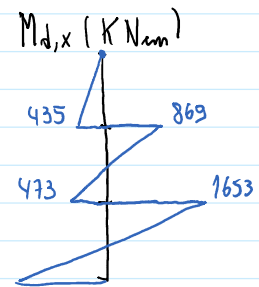
$$M_{sup} = \frac{3209 \cdot 21,78}{21,78 + 43,56 + 135,76} = 348 \text{ kNcm}$$

$$M_{inf} = \frac{3209 \cdot 43,56}{21,78 + 43,56 + 135,76} = 695 \text{ kNcm}$$

2º liga térreo: $n_{sup} = 43,56 \text{ cm}^3$ $n_{inf} = 190,53 \text{ cm}^3$ $n_{vig} = 135,76 \text{ cm}^3$

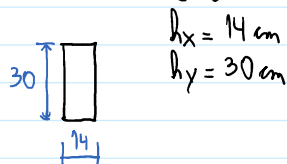
$$M_{sup} = \frac{3209 \cdot 43,56}{43,56 + 190,53 + 135,76} = 378 \text{ kNcm}$$

$$M_{inf} = \frac{3209 \cdot 190,53}{43,56 + 190,53 + 135,76} = 1653 \text{ kNcm}$$

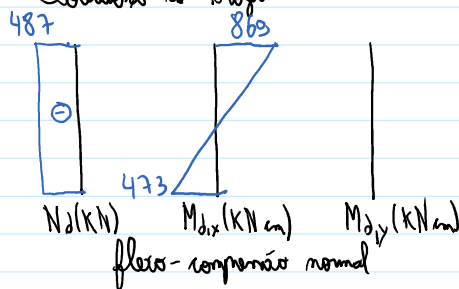


$$M_{inf} = \frac{3209 \cdot 190,53}{43,56 + 190,53 + 135,76} = 1653 \text{ KN cm}$$

Situação de cálculo: prisma reto



Soluição de Projeto



Direção x:

$$e_{1x,A} = \frac{869}{487} = 1,78 \text{ cm}$$

$$e_{1x,B} = \frac{-473}{487} = -0,97 \text{ cm}$$

$$e_{1x,min} = 1,5 + 0,03 \cdot 14 = 1,92 \text{ cm}$$

$$e_{1x,c} \geq \begin{cases} 0,6 \cdot 1,78 + 0,4 \cdot (-0,97) = 0,68 \text{ cm} \\ 0,4 \cdot 1,78 = 0,71 \text{ cm} \end{cases}$$

$$l_{ex} = \min(315; 315 - 40 + 14) = \min(315; 289) = 289 \text{ cm}$$

$$\lambda_x = 346 \cdot \frac{289}{14} = 71,42$$

$$\alpha_{bx} = 1,0 \quad \lambda_{1x} = \frac{25 + 12,5 \cdot 1,78 / 14}{1,0} = 26,59 \quad \lambda_{1x} = 35$$

$$\lambda_x = 71,42 > \lambda_{1x} = 35 \rightarrow \text{esbelto}$$

$$\eta = \min\left(\frac{487}{14 \cdot 30 \cdot 25 / 14}; 0,5\right) = 0,649$$

$$e_{2x,c} = \frac{0,0005}{(0,649 + 0,5)} \cdot \frac{289^2}{14} = 2,60 \text{ cm}$$

Direção y:

$$e_{1y,A} = 0 \text{ cm}$$

$$e_{1y,min} = 1,5 + 0,03 \cdot 30 = 2,4 \text{ cm}$$

$$e_{1y,B} = 0 \text{ cm}$$

$$e_{1y,c} = 0 \text{ cm}$$

$$l_{ey} = \min(315; 315 - 40 + 30) = 305 \text{ cm}$$

$$\lambda_y = 346 \cdot \frac{305}{30} = 35,18$$

$$\alpha_{by} = 1,0 \quad \lambda_{1y} = \frac{25 + 12,5 \cdot 0 / 30}{1,0} = 25 \quad \lambda_{1y} = 35$$

$$\lambda_y = 35,18 > \lambda_{1y} = 35 \rightarrow \text{esbelto}$$

$$\eta = 0,649$$

$$e_{2y,c} = \frac{0,0005}{(0,649 + 0,5)} \cdot \frac{305^2}{30} = 1,35 \text{ cm}$$

Resumo:

$$e_{1x,A} = 1,78 \text{ cm}$$

$$e_{1y,A} = 0 \text{ cm}$$

$$e_{1x,B} = -0,97 \text{ cm}$$

$$e_{1y,B} = 0 \text{ cm}$$

$$e_{1x,c} = 0,71 \text{ cm}$$

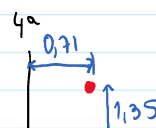
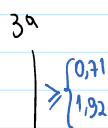
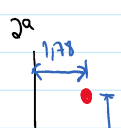
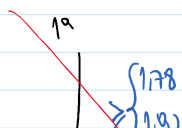
$$e_{1y,c} = 0 \text{ cm}$$

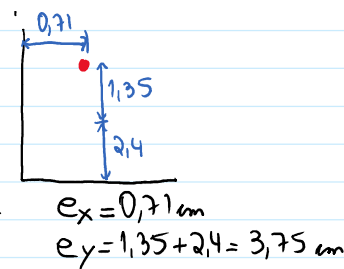
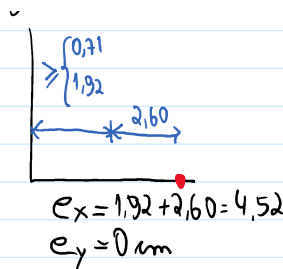
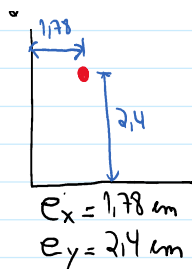
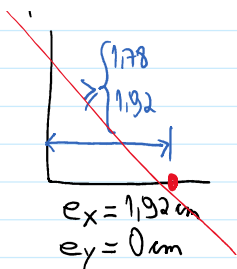
$$e_{1x,min} = 1,92 \text{ cm}$$

$$e_{1y,min} = 2,4 \text{ cm}$$

$$e_{2x,c} = 2,60 \text{ cm}$$

$$e_{2y,c} = 1,35 \text{ cm}$$

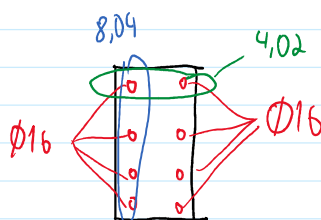
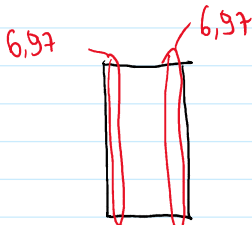




Armaduras: pavimento térreo

Direção x:

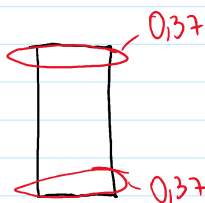
$h = h_x = 14 \text{ cm}$
 $b = h_y = 30 \text{ cm}$
 $d = 10 \text{ cm}$
 $d' = 4 \text{ cm}$
 $M_d = M_{d,xx} = 2202 \text{ kN cm}$
 $N_d = 487 \text{ kN}$



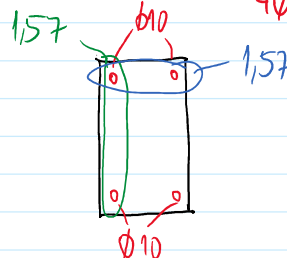
$A_{s, \min} \geq \begin{cases} \frac{0,15 \cdot 487}{43,48} = 1,68 \text{ cm}^2 \\ \frac{0,40}{100} \cdot 14 \cdot 30 = 1,68 \text{ cm}^2 \end{cases}$
 $8\phi 16 \quad 16,08 \text{ cm}^2$

Direção y:

$h = h_y = 30 \text{ cm}$
 $b = h_x = 14 \text{ cm}$
 $d = 26 \text{ cm}$
 $d' = 4 \text{ cm}$
 $M_d = M_{d,yy} = 2020 \text{ kN cm}$



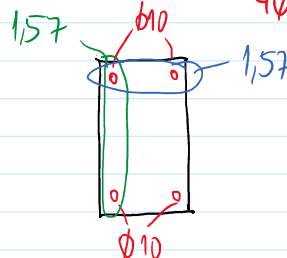
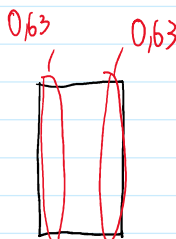
$A_{s, \min} \geq \begin{cases} \frac{0,15 \cdot 222}{43,48} = 0,77 \text{ cm}^2 \\ \frac{0,4}{100} \cdot 14 \cdot 30 = 1,68 \text{ cm}^2 \end{cases}$
 $4\phi 10 \quad 3,14 \text{ cm}^2$



Armaduras: 1º andar

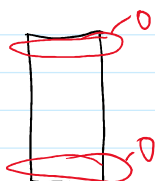
Direção x:

$h = h_x = 14 \text{ cm}$
 $b = h_y = 30 \text{ cm}$
 $d = 10 \text{ cm}$
 $d' = 4 \text{ cm}$
 $M_d = M_{d,xx} = 1164 \text{ kN cm}$
 $N_d = 222 \text{ kN}$



Direção y:

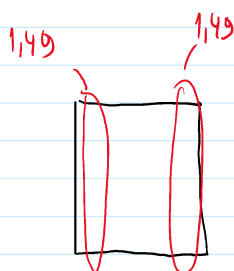
$h = h_y = 30 \text{ cm}$
 $b = h_x = 14 \text{ cm}$
 $d = 26 \text{ cm}$
 $d' = 4 \text{ cm}$
 $M_d = M_{d,yy} = 1020 \text{ kN cm}$
 $N_d = 222 \text{ kN}$



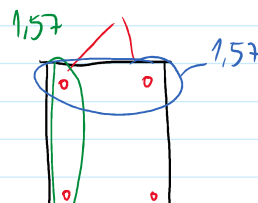
Armaduras: fundações

Direção x:

$h = h_x = 19 \text{ cm}$
 $b = h_y = 30 \text{ cm}$
 $d = 15 \text{ cm}$
 $d' = 4 \text{ cm}$
 $M_d = M_{d,xx} = 2300 \text{ kN cm}$
 $N_d = 601 \text{ kN}$



$A_{s, \min} \geq \begin{cases} \frac{0,15 \cdot 601}{43,48} = 2,07 \text{ cm}^2 \\ \frac{0,40}{100} \cdot 19 \cdot 30 = 2,28 \text{ cm}^2 \end{cases}$
 $\phi 10 \quad 4\phi 10 \quad 3,14 \text{ cm}^2$



$$M_d = M_{d_{ixx}} = 2300 \text{ KN}\cdot\text{cm}$$

$$N_d = 601 \text{ KN}$$

Dimensões y

$$h = h_y = 30 \text{ cm}$$

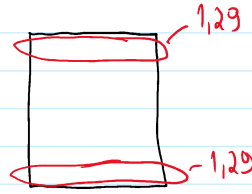
$$b = b_x = 19 \text{ cm}$$

$$d = 26 \text{ cm}$$

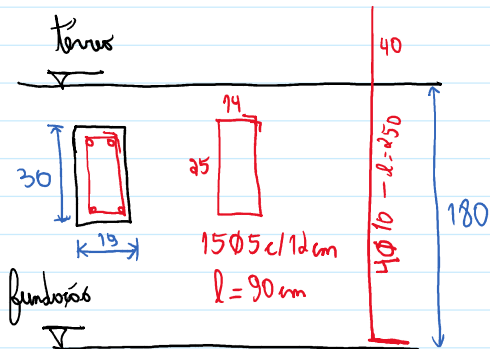
$$d' = 4 \text{ cm}$$

$$M_d = M_{d_{iyy}} = 3700 \text{ KN}\cdot\text{cm}$$

$$N_d = 601 \text{ KN}$$

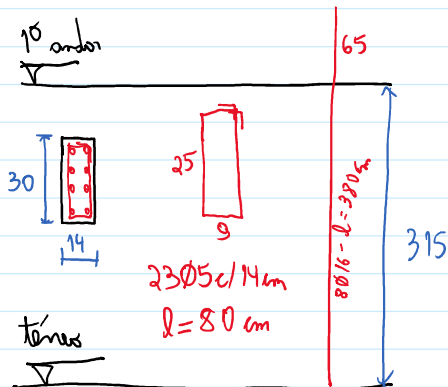


Detalhamentos



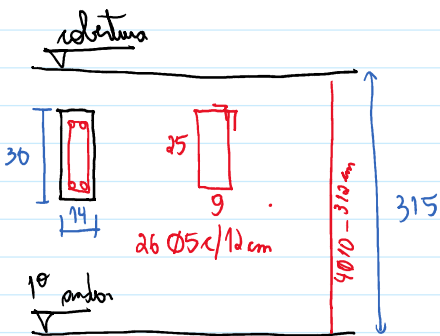
$$\phi_e \geq \begin{cases} 5 \text{ mm} \\ 10/4 = 2,5 \text{ mm} \end{cases}$$

$$s \leq \begin{cases} 20 \text{ cm} \\ 19 \text{ cm} \\ 12 \cdot 1 = 12 \text{ cm} \end{cases}$$



$$\phi_e \geq \begin{cases} 5 \text{ mm} \\ 16/4 = 4 \text{ mm} \end{cases}$$

$$s \leq \begin{cases} 20 \text{ cm} \\ 14 \text{ cm} \\ 12 \cdot 1,6 = 19,2 \end{cases}$$



$$\phi_e \geq \begin{cases} 5 \text{ mm} \\ 10/4 = 2,5 \text{ mm} \end{cases}$$

$$s \leq \begin{cases} 20 \text{ cm} \\ 14 \text{ cm} \\ 12 \cdot 1,0 = 12 \text{ cm} \end{cases}$$

